Preliminary Model Comparison Results

NuclearDetonation.jl: Lagrangian Particle Dispersion

NuclearDetonation.jl

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1 Model Overview

NuclearDetonation.jl is a Lagrangian particle dispersion model for simulating atmospheric transport of radioactive debris from nuclear detonations. Particles are released into a three-dimensional wind field (ERA5 reanalysis on the full 137-level hybrid sigma-pressure grid) and advected forward in time, with sub-grid turbulence, gravitational settling, and dry deposition acting on each particle individually.

Figure 1 shows the release geometry for a representative test case: particles are distributed across a three-layer mushroom cloud structure (lower stem, mid-cap, upper cap) with mass fractions derived from DASA-1251 cloud observations.

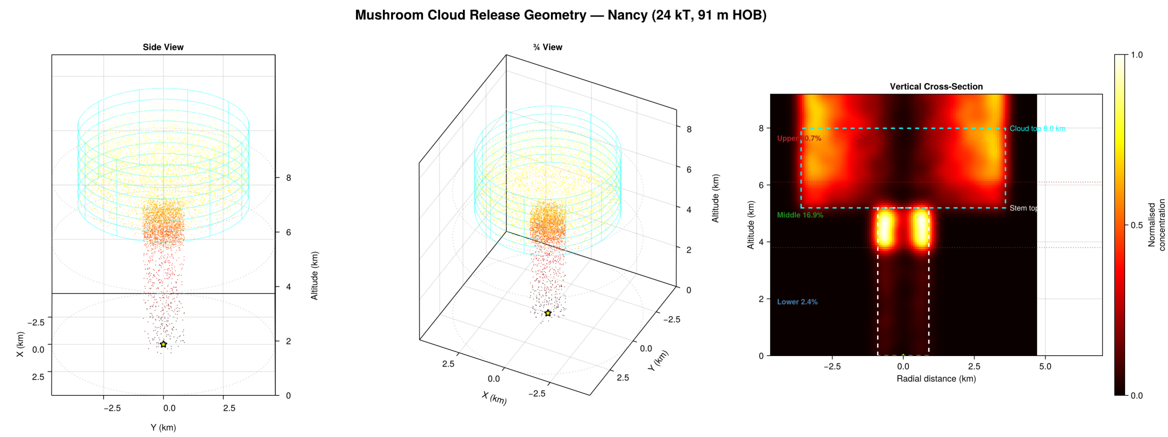

Figure 1: Three-layer mushroom cloud release geometry. Left: 3D particle positions coloured by release layer. Right: kernel density cross-section showing the vertical activity distribution.

1.1 Ornstein–Uhlenbeck Turbulence

Atmospheric turbulence is represented by an Ornstein–Uhlenbeck (OU) stochastic process rather than a simple random walk. In a random walk, each turbulent velocity perturbation is drawn independently at every timestep, producing unrealistically jagged trajectories. The OU process treats the turbulent velocity as a variable with temporal memory: it decays exponentially towards zero with a Lagrangian timescale τ_L, while a stochastic forcing term continually injects energy:

$$u'(t + \Delta t) = u'(t) \exp\!\left(-\Delta t/\tau_L\right) + \sigma_u \sqrt{1 - \exp\!\left(-2\,\Delta t/\tau_L\right)}\ \xi, \tag{1}$$

where σ_u is the turbulent velocity standard deviation and $\xi \sim \mathcal{N}(0, 1)$. The standard deviations σ_u, σ_v, σ_w and timescales τ_L are computed from the Hanna (1982) parameterisation, which uses boundary-layer stability class and height above ground. Separate OU processes are maintained for the horizontal and vertical components.

The practical effect is that particles "remember" their turbulent velocity over timescales of tens of seconds to minutes, producing smoother, more physically plausible dispersion than an uncorrelated walk.

1.2 Gravitational Settling

Each particle is assigned a diameter drawn from a bimodal lognormal distribution (fine and coarse fractions), representing the range of debris sizes from sub-micron fission products to hundreds-of-micron soil particles. The gravitational settling velocity for each particle is computed from Stokes drag with a Cunningham slip correction for small particles:

$$v_g = \frac{\rho_p \, d_p^2 \, g \, C_s}{18 \, \mu} \tag{2}$$

where ρ_p is the particle density, d_p is the diameter, μ is the dynamic viscosity of air (Sutherland's formula, temperature-dependent), and C_s is the Cunningham slip correction factor:

$$C_s = 1 + \frac{2\lambda}{d_p}\left(1.257 + 0.4\,\exp\!\left(-\frac{0.55\,d_p}{\lambda}\right)\right) \tag{3}$$

with λ the mean free path of air. The settling velocity is added to the vertical component of the particle velocity at each timestep, so large particles fall out quickly and deposit close to ground zero whilst fine particles remain aloft and travel long distances.

1.3 Dry Deposition

Surface removal of particles uses a resistance analogy following Zhang et al. (2001). The deposition velocity is:

$$v_d = v_g + \frac{1}{R_a + R_s} \tag{4}$$

where R_a is the aerodynamic resistance (from Monin–Obukhov similarity theory, accounting for atmospheric stability) and R_s is the surface resistance. The surface resistance depends on three collection mechanisms: Brownian diffusion ($E_B \propto \mathrm{Sc}^{-0.54}$), inertial impaction ($E_{\mathrm{IM}}$ via the Stokes number), and interception (E_{IN} via the ratio of particle diameter to collector radius). A rebound correction $R_1 = 1 - \exp(-\sqrt{\mathrm{St}})$ accounts for particles bouncing off surfaces rather than sticking.

When a particle's height drops below a threshold near the surface, its mass is reduced exponentially at a rate $k = v_d/h_{\mathrm{mix}}$, transferring activity to the ground-level deposition field.

2 Nancy 24 kT: Consortium Comparison

The model was first validated against the Upshot-Knothole Nancy test (24 kT, 24 March 1953). Simulated dose rate contours at H+12 hours are compared with digitised observations using the Figure of Merit in Space:

$$\mathrm{FMS} = \frac{A_{\mathrm{obs}} \cap A_{\mathrm{model}}}{A_{\mathrm{obs}} \cup A_{\mathrm{model}}} \tag{5}$$

Six dose rate thresholds are evaluated: 0.4, 1, 4, 10, 40, and 100 mR/h. Each consortium member submitted output in their native format, interpolated onto a common 2 km grid for scoring.

Table 1 summarises the results. Our model achieves a mean FMS of 45.9%, roughly 17 percentage points above the next-best consortium member (UKMO, 28.8%).

Table 1: FMS scores for Nancy 24 kT at H+12.

Dose Rate (mR/h)	Our Model	UKMO	BfS	KIT	DSA/MET	DEMA
0.4	67.1%	60.4%	52.9%	33.8%	45.6%	0.7%
1	58.7%	47.3%	31.1%	27.7%	28.1%	–
4	55.6%	22.4%	18.2%	18.9%	16.9%	–
10	40.1%	15.9%	9.9%	12.1%	9.2%	–
40	28.7%	16.2%	5.4%	4.4%	5.5%	–
100	25.2%	10.5%	2.8%	0.5%	2.2%	–
Mean	**45.9%**	**28.8%**	**20.0%**	**16.2%**	**17.9%**	**0.1%**

3 Side-by-Side Comparisons

The following figures show the digitised observation contours (left panel) and each model's predicted dose rate contours (right panel) for the six thresholds.

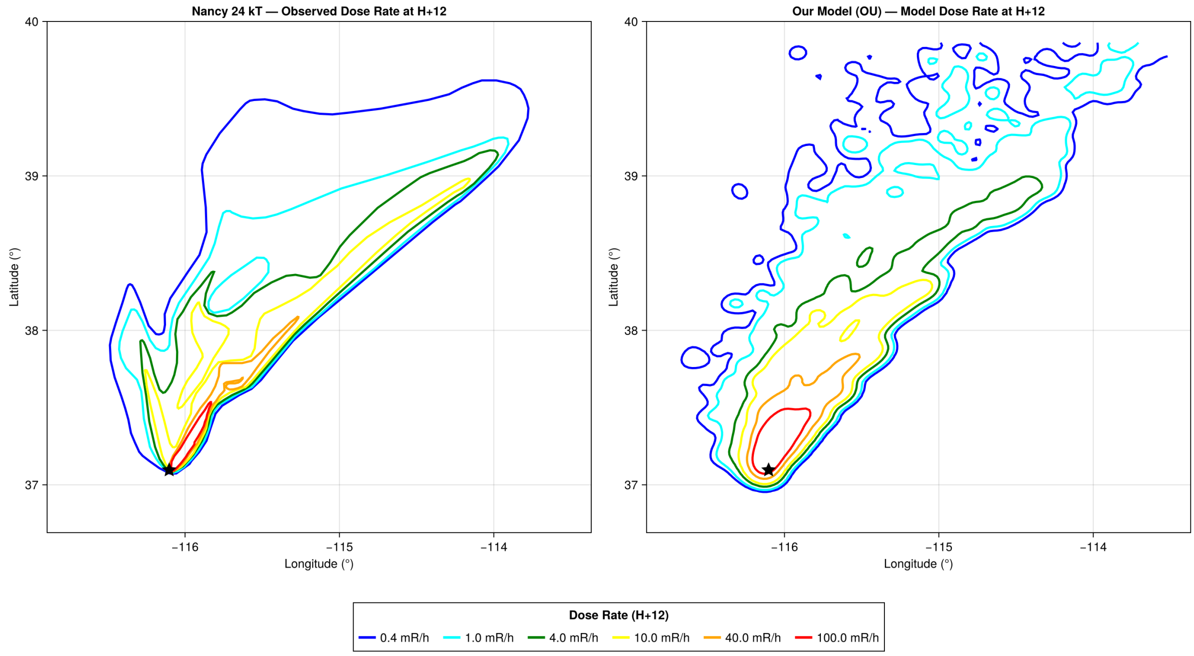

Figure 2: Nancy FMS comparison: our model (right) vs digitised observations (left) for six dose rate thresholds. Mean FMS 45.9%.

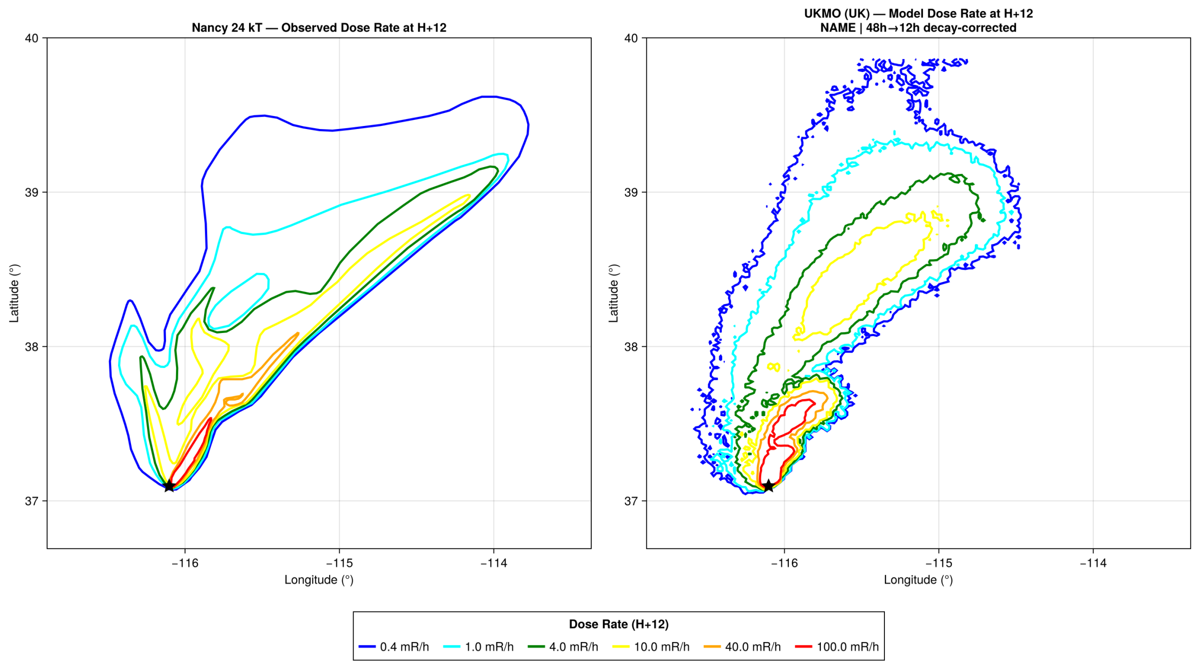

Figure 3: Nancy FMS comparison: UKMO, UK Met Office NAME model (right) vs digitised observations (left). Mean FMS 28.8%.

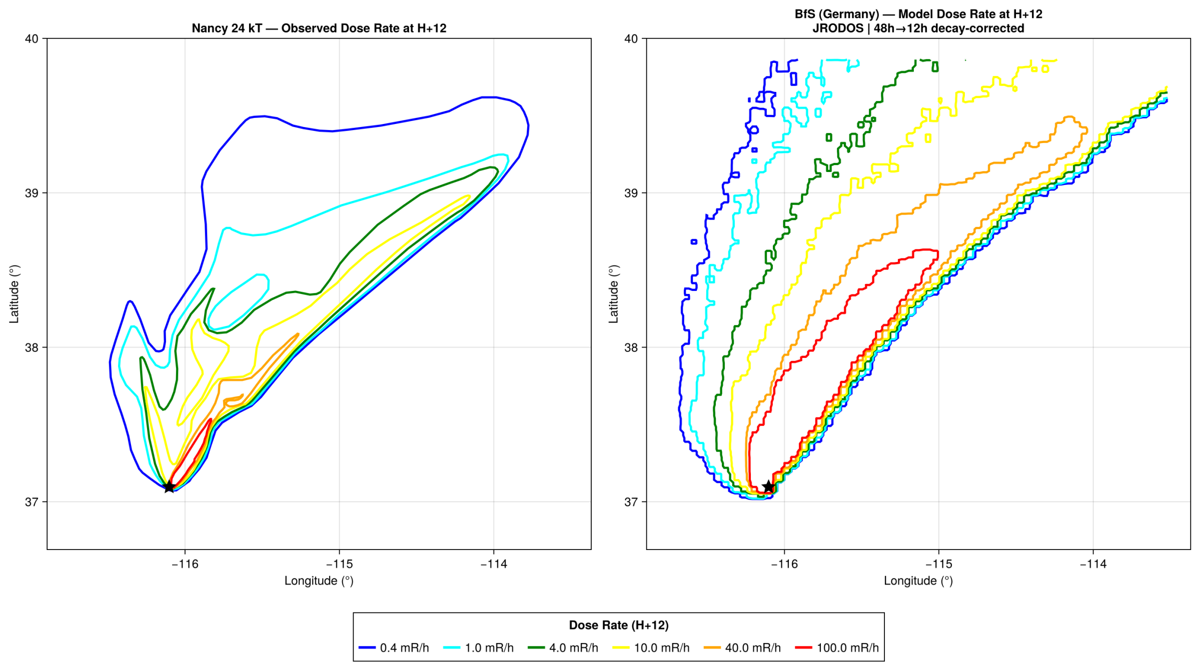

Figure 4: Nancy FMS comparison: BfS, Germany, JRODOS (right) vs digitised observations (left). Mean FMS 20.0%.

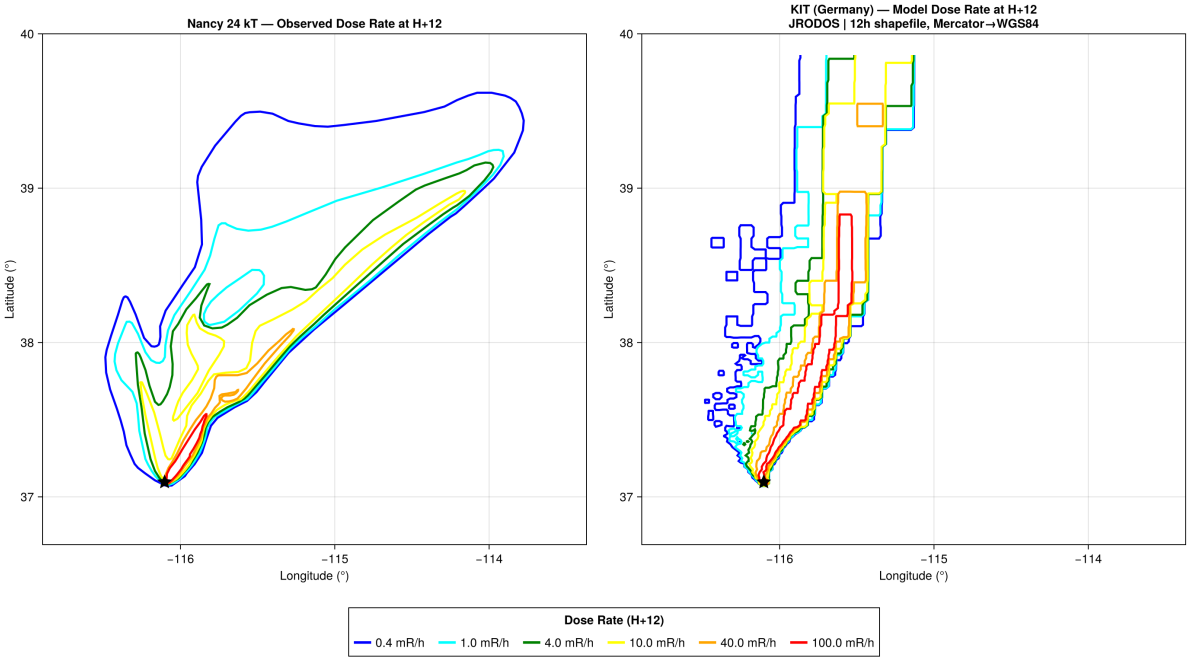

Figure 5: Nancy FMS comparison: KIT, Germany, JRODOS (right) vs digitised observations (left). Mean FMS 16.2%.

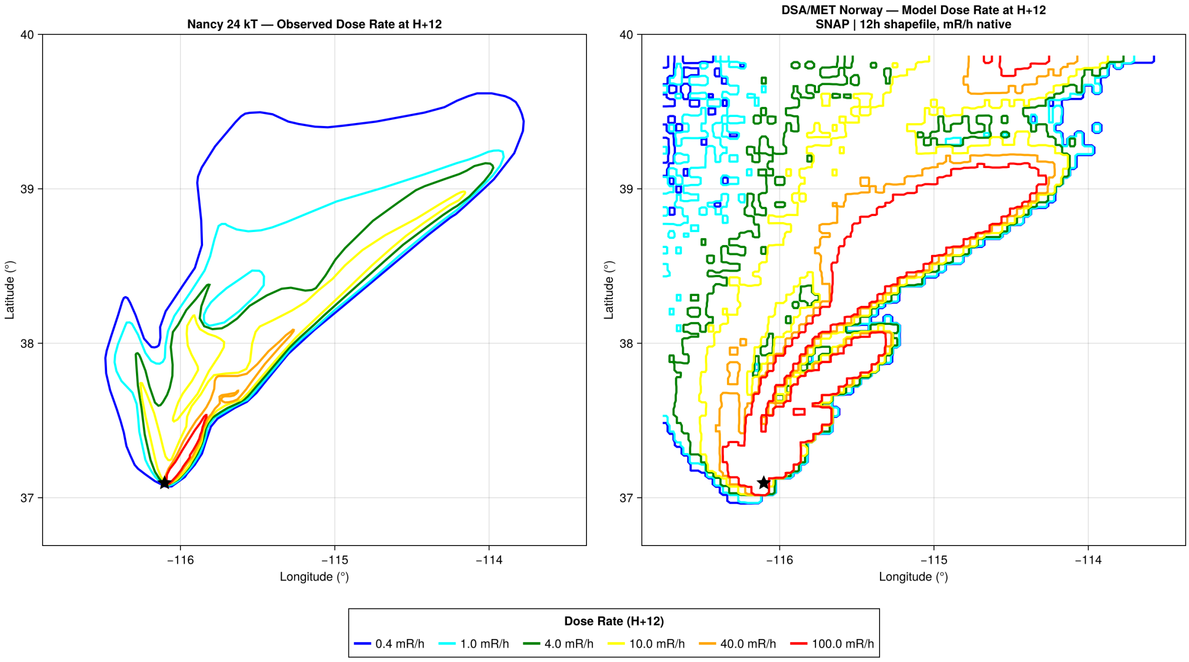

Figure 6: Nancy FMS comparison: DSA/MET Norway, SNAP model (right) vs digitised observations (left). Mean FMS 17.9%.

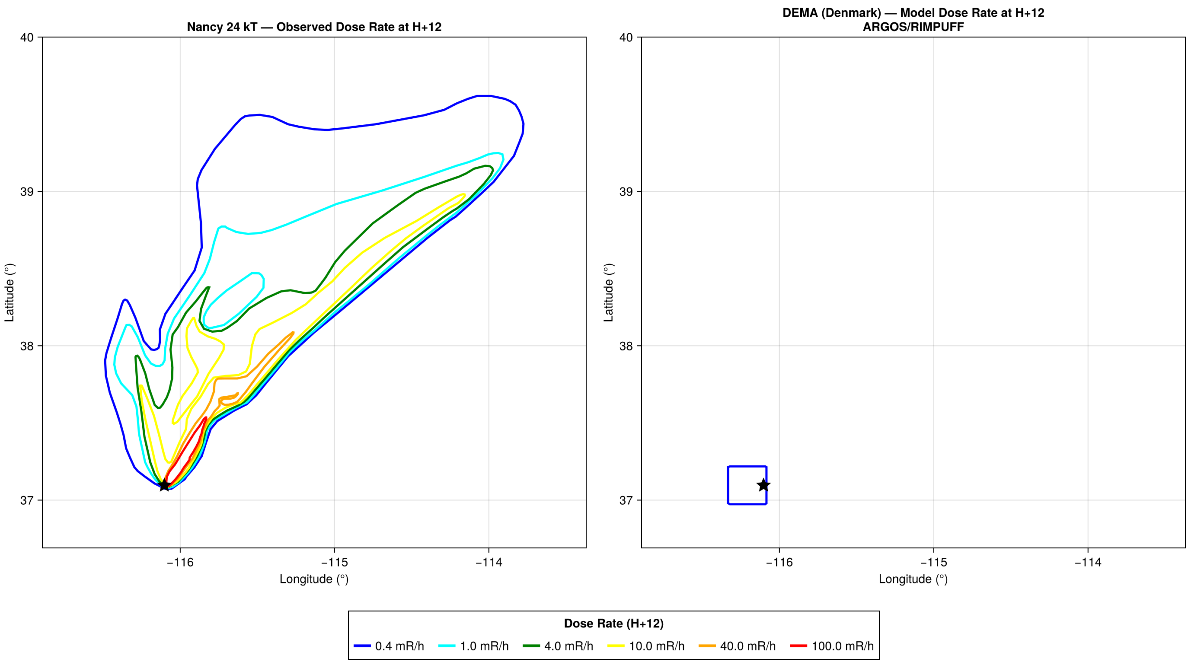

Figure 7: Nancy FMS comparison: DEMA, Denmark, ARGOS/RIMPUFF (right) vs digitised observations (left). Mean FMS 0.1%.

4 Nancy 24 kT: Observed vs Model

Figure 8 shows a direct comparison between digitised observations and model predictions for the Nancy test. The top row compares dose rate contours at H+12; the bottom row compares time-of-arrival (TOA) contours showing when the fallout plume first reached each location. The model TOA is computed from cumulative hourly deposition snapshots, smoothed with a Gaussian kernel and thresholded at 0.1% of the peak to detect the advancing plume front.

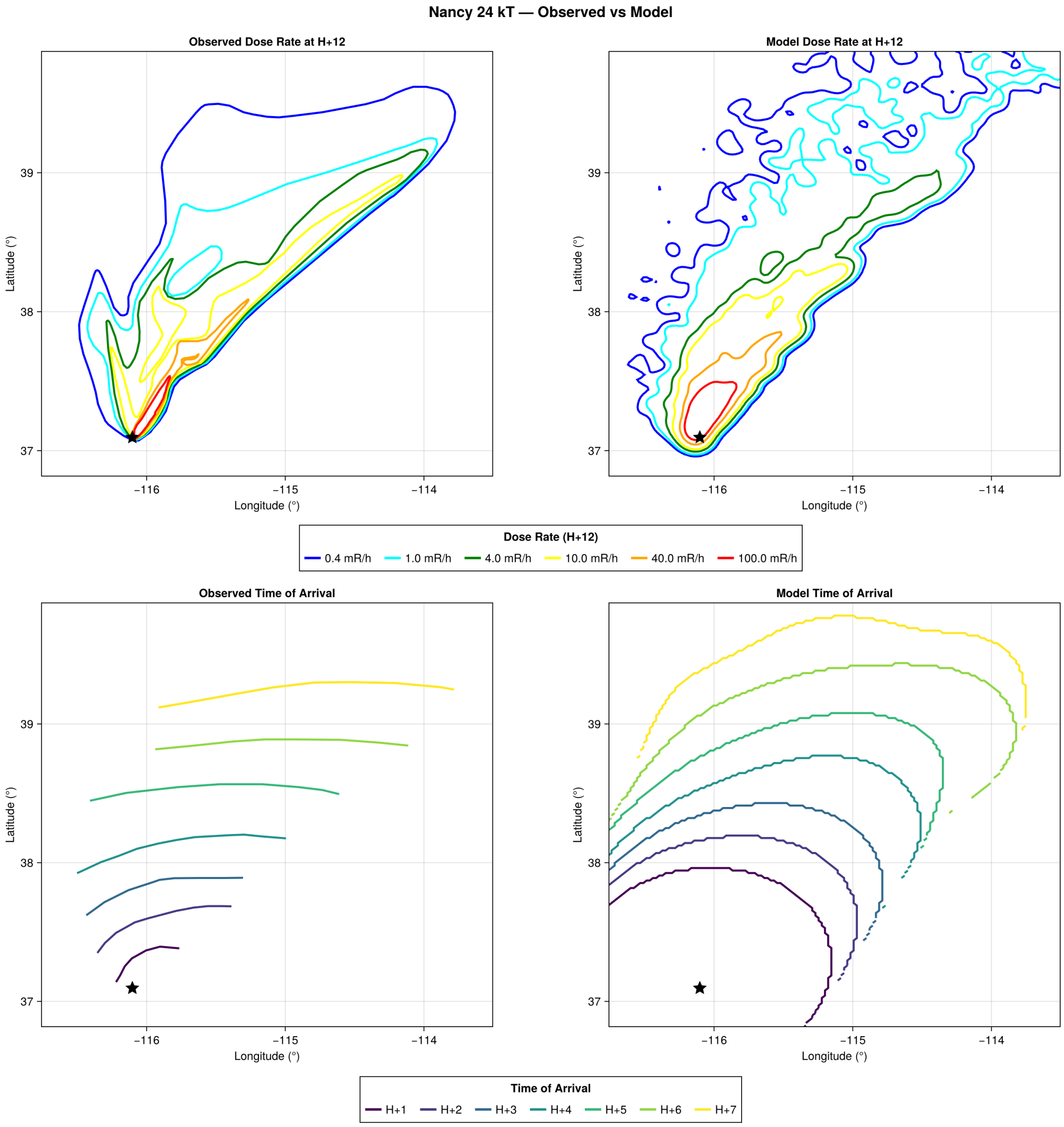

Figure 8: Nancy 24 kT: observed (left) vs model (right). Top row: dose rate contours at H+12 for six thresholds (0.4, 1, 4, 10, 40, 100 mR/h). Bottom row: time-of-arrival contours showing plume progression from H+1 through H+7.

5 Smoky 44 kT: Preliminary Results

The same model and OU turbulence scheme were applied to the Plumbbob Smoky test (44 kT tower shot, 31 August 1957, Nevada Test Site). The three-layer release geometry was initialised from DASA-1251 cloud observations (stem top 10,600 ft, cap centre 22,800 ft, cloud top 35,000 ft MSL), converted to AGL using the NTS surface elevation of 1,409 m. ERA5 reanalysis provides the driving meteorology.

Table 2 presents the per-threshold FMS scores for Smoky. The current best parameter set achieves a geometric-mean FMS of 44.7%, with the modelled plume correctly reproducing the eastward fallout lobe driven by mid-tropospheric westerlies documented in DNA 6004F. Consortium submissions for Smoky are pending; columns are reserved for comparison once results become available. Calibration is ongoing and we expect further improvement as the parameter space is explored.

Table 2: FMS scores for Smoky 44 kT at H+12. Consortium results are pending.

Dose Rate (mR/h)	Our Model	UKMO	BfS	KIT	DSA/MET	DEMA
1	47.4%					
10	44.8%					
50	54.4%					
100	52.5%					
500	40.2%					
1000	32.7%					
Geo. mean	**44.7%**					

Figure 9 compares observed and modelled fields for both dose rate and time of arrival. The dose rate contours (top row) show the ESE plume axis is well captured, whilst the TOA contours (bottom row) demonstrate that the plume advances at the correct speed, with model H+1 through H+12 contours progressing downwind in agreement with the DASA-1251 survey arcs.

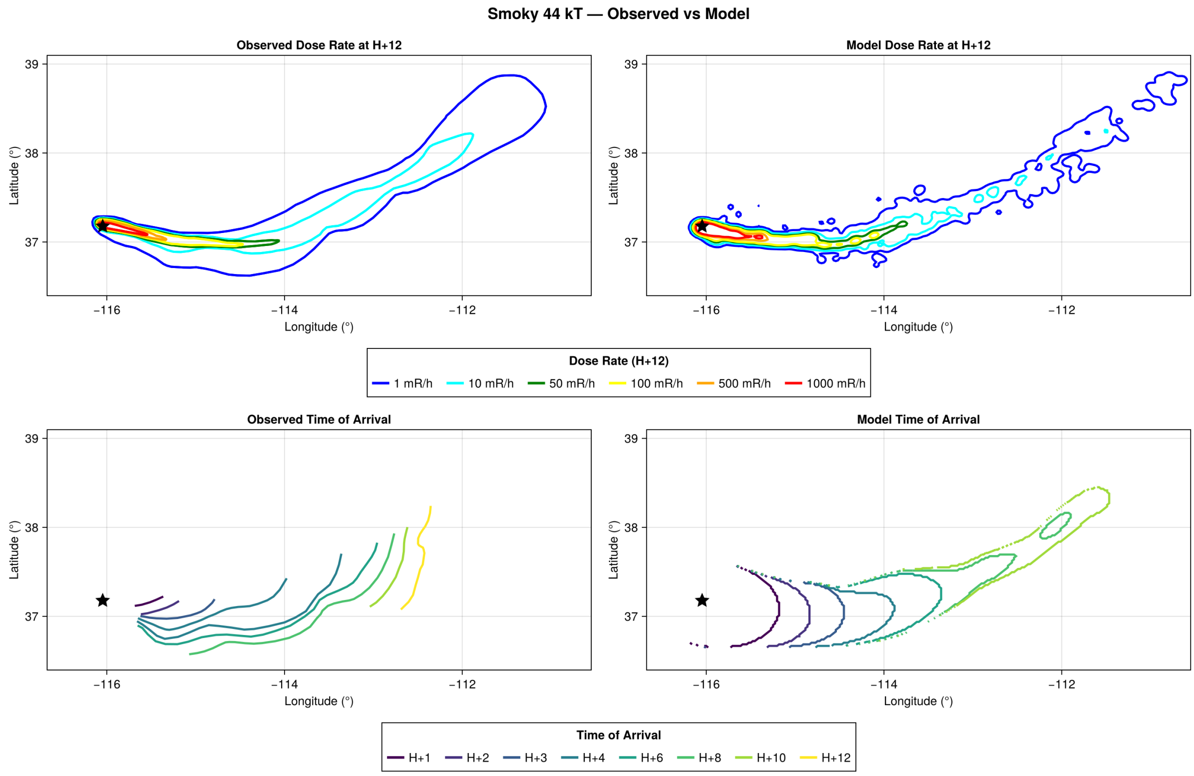

Figure 9: Smoky 44 kT: observed (left) vs model (right). Top row: dose rate contours at H+12 for six thresholds (1, 10, 50, 100, 500, 1000 mR/h). Bottom row: time-of-arrival contours showing plume progression from H+1 through H+12.